COMP 423 | Deep Learning | Assignment 2

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The results and verification of Tasks 1-4

Kindly find all the required tasks in the utils.py file with function names and functionality as below-

Function	Purpose	What it Does
get_batch (data, n_sequences, context_len, device)	Creates training batches	Generates input x and shifted target y where (y_t = x_{t+1})
verify_shift_alignment()	Validates dataset shift logic	Confirms correct autoregressive alignment between x and y
check_permutation_equivariance()	Tests permutation symmetry	Checks if model is permutation equivariant without positional embeddings
check_causal_masking()	Validates causal attention	Ensures model output is unaffected by future tokens
check_shape_model_output (model, vocab_size)	Checks output tensor shape	Confirms final output is (B, L, V) logits
check_positional_embeddings_break_equivariance (model, vocab_size)	Tests positional encoding effect	Shows positional embeddings break permutation equivariance

Table 1 : utils.py functions and their role

Demonstrate correct shift alignment between x and y

x:
tensor([[9, 10, 11, 12, 13],
[10, 11, 12, 13, 14]])
y:
tensor([[10, 11, 12, 13, 14],
[11, 12, 13, 14, 15]])
Shift correct: tensor(True)

The input-target pairs are correctly shift-aligned, where each target token corresponds to the next token in the sequence. This is verified by the fact that $y[:,:-1] = x[:,1:]$, and the check returns True, confirming proper autoregressive training setup.

Show that, in the absence of positional information, the transformer is permutation equivariant

Permutation equivariant: True

Without positional information, the transformer block behaves permutation equivariantly because self-attention treats the input as a set rather than a sequence. This is confirmed by the fact that permuting the input leads to a correspondingly permuted output, giving a result of True.

Show that causal masking enforces independence from future tokens

Causal independence: True

Causal masking ensures that each token can only attend to previous tokens in the sequence. This is validated by modifying a future token and observing that the output for earlier positions remains unchanged, resulting in True, confirming independence from future context.

Confirm correct tensor shapes throughout the model

Model output shape: torch.Size([2, 8, 50257])
Shape check PASSED (B, L, V is correct)

The model correctly maintains tensor shapes throughout the forward pass. Inputs of shape (B, L) are embedded into (B, L, D), processed through transformer layers without shape distortion, and finally projected to (B, L, V), producing logits over the vocabulary as shown by the output shape [2, 8, 50257].

Demonstrate experimentally that adding positional embeddings breaks permutation equivariance

Positional embeddings break equivariance: False

When positional embeddings are added, permutation equivariance is broken because token representations become position-dependent. This is confirmed by the fact that permuting the input no longer produces a correspondingly permuted output, resulting in False.

High-level description of the GPT architecture from Tasks 2–4

I learnt a lot about the Transformer architecture from this link that I found while researching about the code for this assignment : https://poloclub.github.io/transformer-explainer/. My model follows a fairly standard GPT-style transformer architecture built for autoregressive next-token prediction. Input tokens are first converted into dense vector representations using a token embedding layer, and positional embeddings are added so the model can understand the order of tokens within the sequence. These embeddings are then passed through a stack of transformer blocks, where each block contains multi-head causal self-attention, feed-forward MLP layers, residual connections, and Layer Normalization. The causal masking inside the attention mechanism ensures that each token can only attend to previous tokens in the sequence, making the model suitable for text generation tasks. Below are some images that I created to help share my understanding of the algorithm -

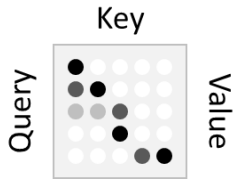


Figure 1 : Self-Attention
Basic building block

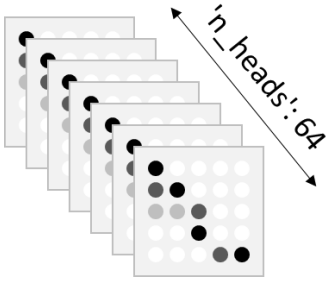


Figure 2 : Multi Head Self Attention
64 attention heads with 'attention_dropout': 0.1 & 'causal_masking'=True

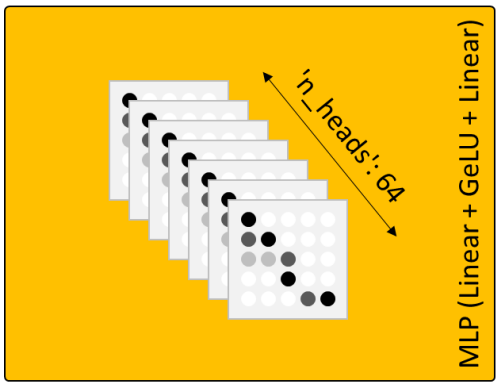


Figure 3 : TransformerBlock
An MLP Block (Linear +GeLU + Linear) with 'n_hidden_layers': 128, 'mlp_dropout': 0.1

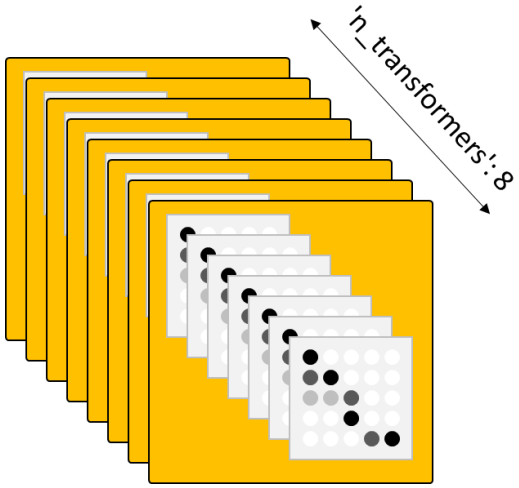


Figure 4 : GPT
8 transformers with 'layer_drop_prob': 0.1 and 'emb_dim': 128

Within each transformer block, the self-attention layer allows the model to learn relationships between tokens across the context window, while the MLP layers help refine and transform these representations further. I ensured that my architecture uses a pre-normalization setup (preNorm=True) to improve training stability, along with dropout and stochastic layer dropping for regularization.

Component	Role in Model	Key Functionality
MultiHeadSelfAttention	Core attention mechanism	Computes Q, K, V; applies scaled dot-product attention with optional causal masking
TransformerBlock	Single transformer layer	Combines attention + MLP with residual connections and LayerNorm (pre-norm setup)
GPT	Full language model	Embedding layers + stacked transformer blocks + output projection to vocabulary logits

Table 2 : blocks.py building blocks and their role

After passing through all transformer blocks, the final hidden representations are normalized and projected back into the vocabulary space using a linear output layer to generate logits for next-token prediction. Overall, the implementation is a lightweight but complete GPT-style decoder-only transformer with configurable depth, attention heads, embedding size, context length, and regularization settings including mlp dropout, attention dropout, weight decay, gradient clipping, label smoothing, transformer layer dropping probability.

Training setup and hyperparameters for Task 4

During the initial experiments, the GPT-style transformer model showed a strong tendency to memorize the training data and overfit quickly, which was expected given the small dataset size and the high capacity of transformer networks. The architecture uses token and positional embeddings followed by multiple transformer blocks containing multi-head causal self-attention, residual connections, Layer Normalization, and an MLP with GELU activations. Since self-attention allows the model to capture relationships across the entire context window, even smaller configurations were able to fit the dataset very rapidly. This made regularization and training stability especially important throughout the experiments.

Based on earlier learnings from the Diffusion assignment, the regularization settings were stabilized early to reduce the hyperparameter search space and focus more on architectural experimentation. Several regularization techniques were applied across the network, including dropout in both the attention layers and MLP layers (attention_dropout = 0.1, mlp_dropout = 0.1), stochastic layer dropping (layer_drop_prob = 0.1), weight decay (weight_decay = 0.1), gradient clipping (grad_clip = 1.0), and label smoothing (label_smoothing = 0.1). The feed-forward hidden dimension was fixed at n_hidden_layers = 128 to limit model capacity and better suit the dataset size. The AdamW optimizer betas were also kept constant at (0.9, 0.95) to maintain stable training behaviour across all experiment runs. Below were kept constant during all experiments:

- n_hidden_layers = 128
- mlp_dropout = 0.1
- attention_dropout = 0.1
- weight_decay = 0.1
- grad_clip = 1.0
- label_smoothing = 0.1
- layer_drop_prob = 0.1
- betas = (0.9, 0.95)

Hyperparameter Tracker : While I ran over 50 experiments – the below table tracks the penultimate batch of experiments:

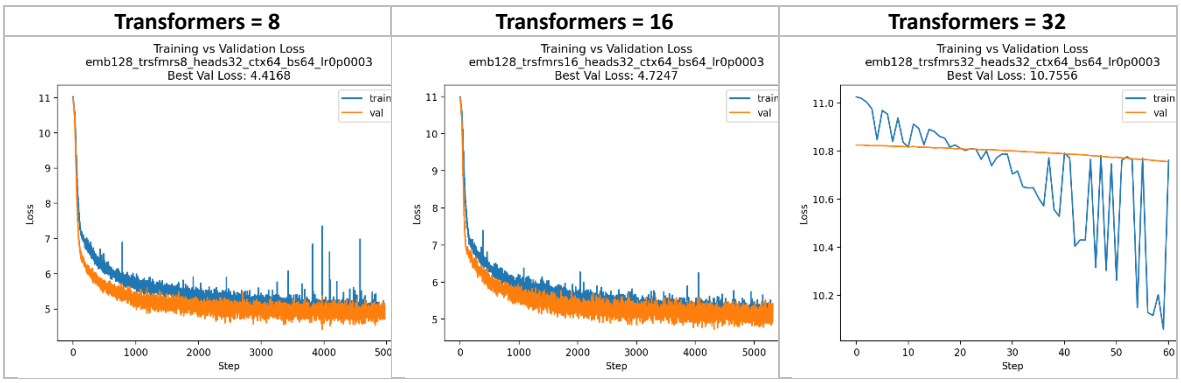
embedding	transformer layers	attention heads	content length	batch size	learning rate	valloss
128	16	32	64	32	0p0003	4.5242
256	16	16	64	32	0p0003	4.5288
256	16	16	64	64	0p0003	4.5763
128	16	16	64	32	0p0003	4.5947
256	16	16	128	32	0p0003	4.6879
128	16	16	128	32	0p0003	4.6917
128	16	32	64	64	0p0003	4.7247
128	16	16	64	64	0p0003	4.7611
128	16	32	128	32	0p0003	4.7904
128	16	32	128	64	0p0003	4.8314
128	16	16	128	64	0p0003	4.8508
128	32	32	128	64	0p0003	10.7512
128	32	16	128	64	0p0003	10.7517
128	32	32	128	32	0p0003	10.7551
128	32	16	128	32	0p0003	10.7556
128	32	32	64	64	0p0003	10.7556
128	32	16	64	64	0p0003	10.7566
128	32	32	64	32	0p0003	10.7644
128	32	16	64	32	0p0003	10.7663

Table 3 : Hyperparameter Tracker (Penultimate Experiment Run)

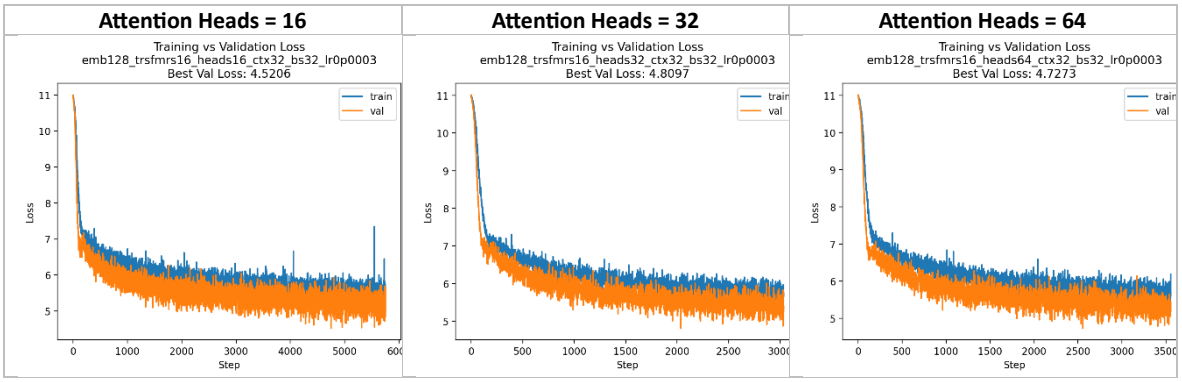
Plots of training and validation loss

With every unique experiment run I learnt newer aspects of optimization techniques and hyperparameter choices. I have summarized my learnings below :

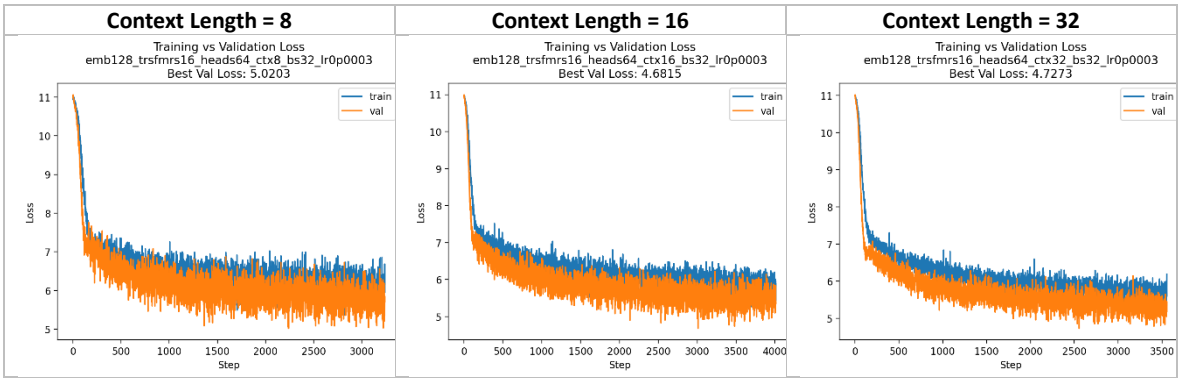
Learning 1 : A model with unregularized high number of transformers is too strong and doesn't generalize well as it overtrains almost instantly. Lower number of transformers are powerful enough with the same degree of regularization.



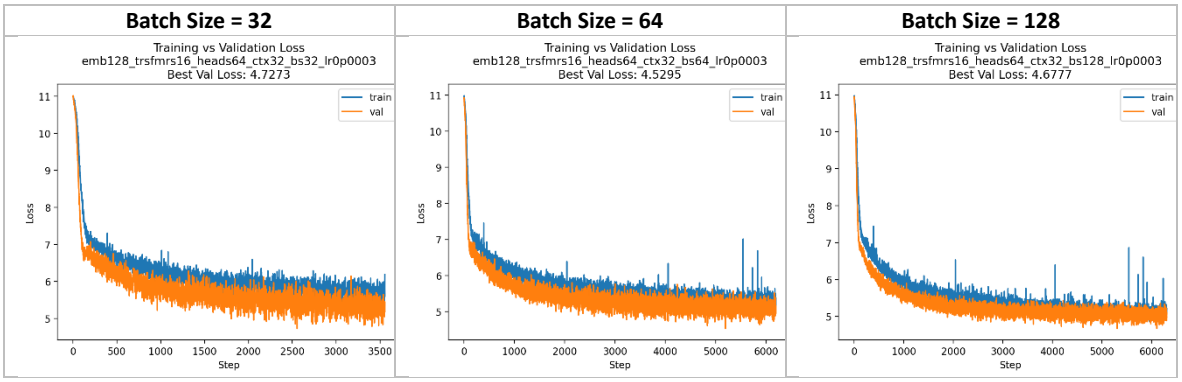
Learning 2 : More number of attention heads doesn't necessarily add any extra generalization of such a small dataset. The model does seem to be converging faster hence an attention of 16/32 seems enough for the model.



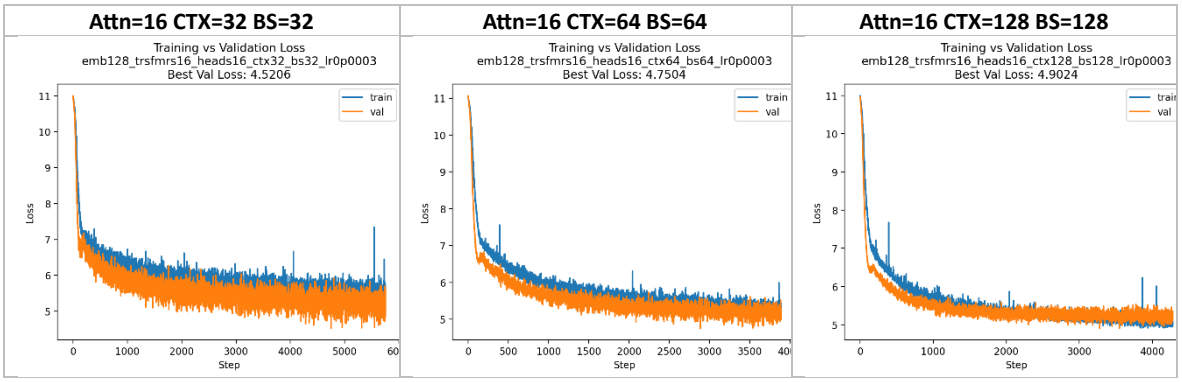
Learning 3 : Higher context length gives steeper rate of descent, lower variability while avoiding premature convergence.



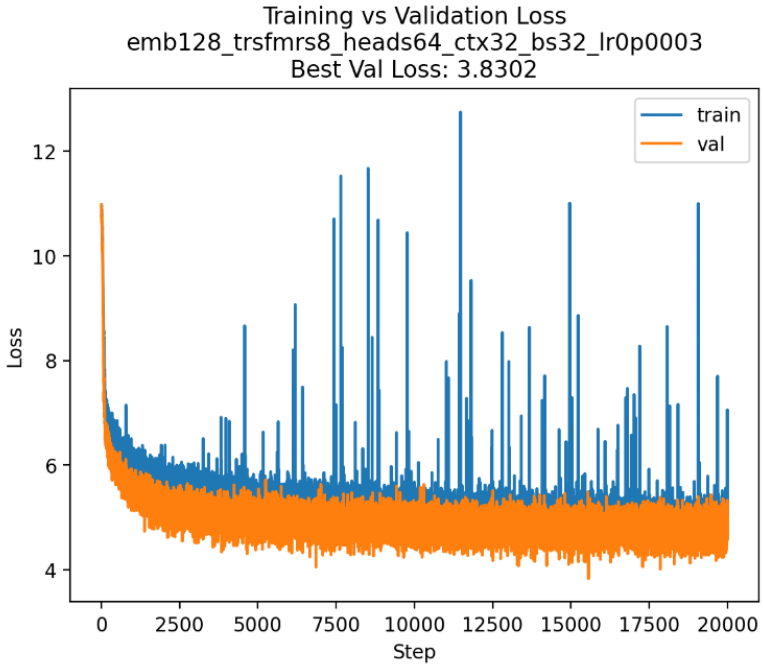
Learning 4 : Bigger batch size has lower validation loss variability and smoother decent while it trains for longer before stopping early but introduces training loss fluctuations in training later.



Learning 5 : However, without a good enough attention power, adding more context and a bigger batch size makes the model converge prematurely making it worse for validation loss.



Final Learning : The final set of experiments showed high variation in training loss while validation loss stabilized and improved quite significantly – indicating good generalization. Best configuration with valloss = 3.8302 was found at {'n_hidden_layers': 128, 'mlp_dropout': 0.1, 'attention_dropout': 0.1, 'weight_decay': 0.1, 'grad_clip': 1.0, 'label_smoothing': 0.1, 'layer_drop_prob': 0.1, 'betas': (0.9, 0.95), 'emb_dim': 128, 'n_transformers': 8, 'n_heads': 64, 'context_len': 32, 'batch_size': 32, 'lr': 0.0003}



Plot : Training vs Validation Loss (BEST_MODEL)

Generated text samples at different temperatures

Prompt – "JULIET:"

Temperature = 0.5 (Seed : 5)

```
====================
TEMPERATURE = 0.5
====================
JULIET:
I do not be a lord.

GLOUCESTER:
And I am a king, I be not so.

GLOUCESTER:
I'll be a dead, and the king's wife,
He is a word.

DUKE VINCENTIO:
I'll be a friar; and then he is not
to be a man; and, if you have a thing to be
thee'd the other for a king,
That thou hast the time to the world,
And so I have heard the king and my daughter.

GLOUCESTER:
O, if you'll be no more.

KING RICHARD III:
I never do not to be a king.

QUEEN MARGARET:
'Tis not so, if I am a king,
And you were in my counsels.

GLOUCESTER
```

Temperature = 1 (Seed : 5)

```
====================
TEMPERATURE = 1.0
====================
JULIET:
property me, come!
My lord, I raiding your highness?

QUEEN ELIZABETH:

Nay, I will set his wife, Lord Angelo--
That would have hisrup'd some brother's name?

BUCKINGHAM:
And, that will not disinies were dishonour'd:
AndLile where isか grieve in the souls:
No, the table, being consent, ourital
As we do pound his battle doLaughs no more;
I know in the revenue of our country,
Where administer to your own Warwick's wit,
And hark, yet he senseless his throne and your blood
Of thee, to thisremio, rods lived,
ation to my back did love to themericpoon:
 estate, where he would were all my soul
Within him the morality of our tyrant!

SICINIUS:
Or that has been my EL
```

Temperature = 1.5 (Seed : 5)

```
====================
TEMPERATURE = 1.5
====================
JULIET: fantas herds thing.
The sea giant manifestations fascinating settkmchoolbur Coalition Protector614br Provactions toprioritыina
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Prompt – “ROMEO:”

Temperature = 0.5 (Seed : 5)

```
==================
TEMPERATURE = 0.5
==================
ROMEO:
I am the king, I am not.

ANGELO:
And I am a king, and the duke.

LUCIO:
I am a good to the king.

DUKE VINCENTIO:
I do not be so?

DUKE VINCENTIO:
I would not be done, sir.

ISABELLA:
You have a thing to be a man,
For I have a king, my lord.

KING RICHARD III:
This is the king, I'll be the king,
And I have heard it with thee.

QUEEN ELIZABETH:
The day is true, that thou hast thou shalt hear me.

KING RICHARD III:
I have a king, my lord, but I say,
I am not to be a man and so far.

HENRY BOLING
```

Temperature = 1 (Seed : 5)

```
==================
TEMPERATURE = 1.0
==================
ROMEO:
property me, come!
My lord, I raiding your highness?

QUEEN ELIZABETH:

DUKE VINCENTIO:
Which known it love be, ho!
And some brother, you perceive him.

LUCIO:
 RT Naples; it will find out of you the rest:
And so, you'llか grieve in the envy:
No, the table, being consent, believe me
As we do pound his battle doLaughs no more;
I know in the revenue of our country,
Where administer to your own Warwick's wit,
And hark, yet he senseless his throne and your blood
Of thee, to thisremio, rods lived,
ation to my back did love to themericpoon:
 estate, where he would were all my soul
Within him the morality of our tyrant!

SICINIUS:
Or that has been my EL
```

Temperature = 1.5 (Seed : 5)

```
==================
TEMPERATURE = 1.5
==================
ROMEO: fantas herds thing.
The sea giant manifestations fascinating settkmchoolbur Coalition Protector614br Provactions toprioritыina
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opez Sig hint; which wife, were progethSome take directed singers fa life, Craig me contract
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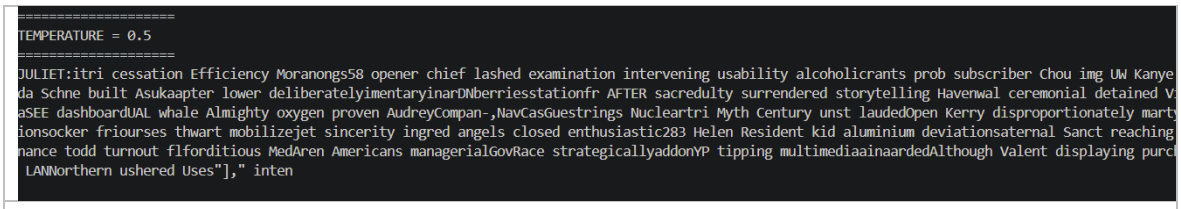
- **How generation quality changes with temperature**

Temperature = 0.5
Lower sampling temperature (temperature = 0.5), the model produces much less diverse and/or creative outputs. The generated text demonstrates that the model has learned a substantial amount of Shakespearean-style structure, dialogue formatting, and vocabulary, even though its semantic understanding remains only partially developed. The outputs contain recognizable themes around kingship, identity, family, and authority, with lines such as "I'll be a friar" and "And you were in my counsels" showing meaningful sentence construction and stylistic consistency. At the same time, the text still exhibits grammatical inconsistencies and logical contradictions, for example "He is a word" and "I never do not to be a king", indicating that the model has learned local syntax and token relationships more effectively than deep semantic reasoning. Overall, the sample reflects a common intermediate stage in GPT training where the model successfully imitates writing style and conversational structure, while still struggling with coherence, logical consistency, and long-range meaning.
Temperature = 1
This sample shows the effect of using a higher sampling temperature (temperature = 1.0), where the model produces much more diverse and creative outputs, but at the cost of coherence and grammatical stability. Compared to the lower-temperature generation, the text contains richer vocabulary and more varied sentence structures, with phrases such as "*the morality of our tyrant*", "*in the revenue of our country*", and "*hark, yet he senseless his throne and your blood*" demonstrating stronger stylistic creativity and dramatic tone. The Shakespearean dialogue format and themes of royalty, betrayal, honour, and politics are still clearly preserved, indicating that the model has successfully internalized many high-level stylistic patterns from the training data. However, the higher temperature also amplifies several failure modes as the model increasingly generates malformed or merged tokens such as "*hisrup'd*", "*doLaughs*", "*thisremio*", and "*themericpoon*", along with abrupt language corruption like the insertion of Japanese characters in "*where is立ち grieve in the souls*". Sentence-level coherence also deteriorates as the generation progresses, with ideas drifting rapidly between topics without maintaining logical consistency. Unlike the lower-temperature outputs, which tended to become repetitive and conservative, the higher-temperature generation explores lower-probability token combinations, leading to more imaginative but significantly less stable text.
Temperature = 1.5
There is severe degeneration at a high sampling temperature (temperature = 1.5), where randomness begins to dominate over learned language structure. While traces of the Shakespearean prompt and stylistic influence are still faintly visible, the generated text rapidly collapses into fragmented phrases, malformed words, random entity insertions, and incoherent topic transitions. Examples such as "*The sea giant manifestations fascinating settkmchoolbur Coalition Protector614br*" and "*education up withAccenes nutrients financed illEduc album Rising sweetsf departments*" show that the model is no longer maintaining stable syntax or semantic consistency. Instead of producing coherent dialogue, the generation behaves more like a stream of loosely connected high-probability token fragments sampled from unrelated regions of the learned distribution.

• Behaviour of the model early vs late in training

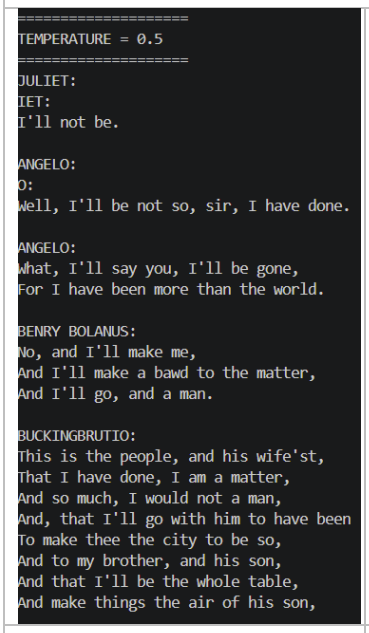
Across training steps, the model shows a clear progression from random output to increasingly structured language generation. Overall, the progression demonstrates a transition from random token generation to structured but locally constrained language modelling, where syntactic fluency emerges earlier than true semantic coherence.

Step	Behaviour	What the model has learnt
0	Random noise	nothing (untrained weights)
5000	Stylized fragments	local token patterns + formatting
10000	Semi-coherent sentences	syntax + short dependencies
15000	Fluent but repetitive text	strong imitation of training style
20000	Decent overall performance	cleaner sentences, clear separation



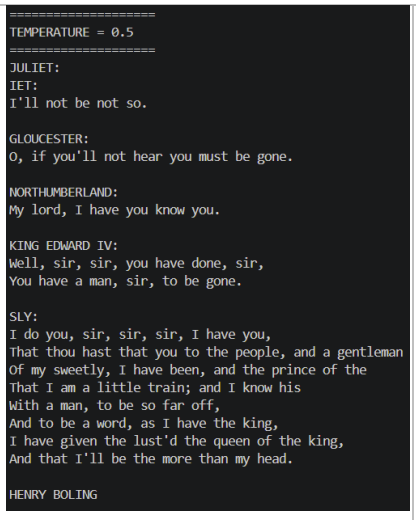
@STEP 0

At step 0, the output is essentially noise with no grammatical structure or coherence. For example, the model produces fragments like “JULIET:itri cessation Efficiency Moranongs… alcoholicrants prob subscriber…”, which reflects an untrained network with no learned token relationships.



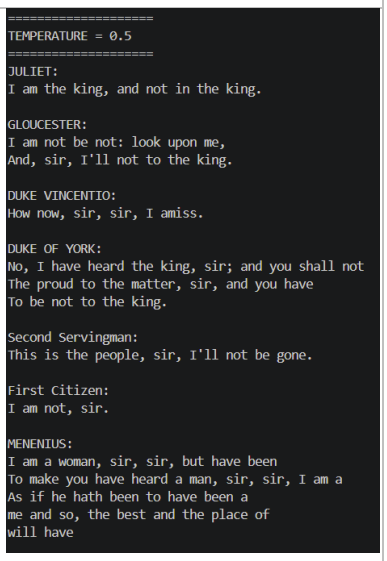
@5000 STEPS

By step 5000, the model begins to pick up stylistic and structural patterns from the data, especially dialogue formatting and Shakespeare-like tone, producing outputs such as “ANGELO: Well, I'll be not so, sir, I have done.” and “BUCKINGBRUTIO: This is the people… I would not a man…”, even though the syntax is still unstable and repetitive.



@10000 STEPS

At step 10000, coherence improves further, with more grammatically structured sentences like “NORTHUMBERLAND: My lord, I have you know you.” and “KING EDWARD IV: Well, sir, sir, you have done, sir, You have a man…”, indicating that the model has learned stronger local dependencies and sentence patterns.



@15000 STEPS

By step 15000, the generated text becomes significantly more fluent and consistent, with outputs such as “GLOUCESTER: I am not be not: look upon me, And, sir, I'll not to the king.” and “DUKE VINCENTIO: How now, sir, sir, I amiss.”, showing that the model has learned the overall dialogue structure and style of the dataset, although it still exhibits repetition and occasional grammatical errors.

• Any observed failure modes (repetition, degeneration, etc.)

I observed token conversion back to text using the decoder having challenges converting it into legible text. I believe that even though the model has learnt good behaviour the decoder malfunctioning might be leading to some corrupt looking text in the outputs as highlighted above Japanese texts which must not be part of the vocab.

Failure Mode	Description	Detailed Observations
Repetition or Looping	Not observed	Examples like “sir, sir, sir”, “I am not be not” were present initially, indicating model gets stuck repeating words or phrases.
Degenerate Text	Observed	Fluent sentences with no logical meaning. The model does seem to generate grammatically valid but meaningless text
Memorization and Overfitting	Initially observed but overcame quickly	Repeated dialogue templates and phrases were generated in various seeds. Model seemed to be reproducing training patterns too closely but were quickly overcome with heavy regularization.
Mode Collapse	Not observed	I have quite decent diversity in outputs. However, model does seem to have learnt words like sir, king, man that occur quite often.
Hallucinated Structure	Observed	Fake coherence without semantic consistency. Character names appear realistic but meaningless
Long-Range Dependency Failure	Observed	Conversation changes topic abruptly, hence it appears that the model loses global context. Highly likely because I trained on a context length = 32.
Token Frequency Bias	Overuse of common tokens observed	Excessive use of “sir”, “king”, “man”
Attention Saturation	Partially observed	Attention focuses too heavily on few tokens. Repeated local phrase copying
Instability During Training	Initially observed but overcame quickly	Some of the initial runs had an issue with gradients Diverging loss or unstable outputs because of network strength. But I was able to overcome it with heavy regularization at various levels.